# Sameer Jain

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## Education

**Georgia Institute of Technology**: Atlanta, GA — May 2028
Bachelor of Science in Industrial Engineering
Concentration in **Probability and Statistics, Data Analytics, and Machine Learning** — GPA: 3.83

## Relevant Experience

**AT&T** | *Data Science Engineer, Atlanta, GA* — June 2026 – August 2026

- Built an agentic AI to automate SQL generation and Snowflake Cortex ML forecasting, reducing manual input by 500+ hours annually by using Copilot Studio to orchestrate workflows across 25M+ financial records.
- Automated ingestion of 25M+ Azure SQL Analytics records into Snowflake via Power Automate + Excel Office Scripts, converting raw financial data into processable datasets and eliminating 100% of manual input.
- Built 20+ Excel Office Scripts to automate financial statement standardization and KPI reporting, saving 350+ analyst hours annually by using automated feature engineering and workflows across 150+ recurring reports.
- Leveraged Snowflake + Cortex ML (forecasting, anomaly detection, embeddings, time-series models) to model revenue and cash flow, detecting 1,000+ anomalies and improving reporting efficiency by 70%.

**Georgia Institute of Technology** | *Undergraduate Teaching Assistant, Atlanta, GA* — May 2025 – Present

- Instructed 400+ students in Pandas, NumPy, SQL, BeautifulSoup, Regex, API integration, and Scikit-Learn.
- Guided 400+ students on end-to-end analytics projects, including ETL pipelines, and API integration.
- Reviewed 30,000+ lines of code delivering insights on 200+ gigabyte datasets exceeding 500M rows.
- Automated grading for 150+ coding homework problems and spent 180+ hours of individual student tutoring.

**Georgia Tech Energy Forecasting Lab** | *Machine Learning Researcher, Atlanta, GA* — Aug 2025 – Dec 2025

- Built PyTorch models to predict energy use in Atlanta for 1,200+ buildings from 20+ climate/material features.
- Achieved sub 10% MAE and 80% faster runtimes, reducing multi-hour simulations to minutes per scenario.
- Built 3+ rapid urban energy models to support a peer-reviewed sustainable design publication.
- Developed NASA ETL pipelines for satellite data using GeoPandas, Rasterio, and GDAL on 1200+ buildings.

## Selected Projects

**Brain Tumor MRI Classification** | *Machine Learning* — January 2026 – May 2026

- Compared supervised learning, clustering, and anomaly detection across 6,056 MRI crops using PCA, classical classifiers, ResNet50 embeddings, and convolutional autoencoders.
- Aggregated 2,048-dimensional ResNet50 features into 603 image-level vectors; K-Means achieved an adjusted Rand index of 1.00 on the single-source dataset.
- Built a glioma-only autoencoder with 100-component PCA and Mahalanobis scoring; evaluated 4,452 scans with held-out glioma images and a threshold derived exclusively from training data.

**Systematic ETF Research Engine** | *Data Science & Engineering* — August 2026 – Present

- Built a Python/DuckDB pipeline for 100 ETFs and 291K+ market observations, with versioned Parquet datasets, point-in-time features, and purged expanding-window validation.
- Froze an 18-feature Elastic Net before evaluating 133 held-out weeks; achieved 0.130 mean Spearman information coefficient, with dependence-adjusted significance (one-sided HAC $p = 0.009$).
- Implemented append-only forward paper predictions with deterministic hashes and delayed evaluation after 21 SPY sessions, preserving immutable forecasts for prospective assessment.

**Granted – JPMorgan Chase Data for Good Hackathon** | *Data Scientist* — October 2026

- Co-built Granted, winning 1st place in the AnnieCannons challenge; developed a React/FastAPI prototype for nonprofit grant analytics, opportunity matching, and AI-assisted editing.
- Engineered versioned Parquet datasets from 304 grant records and 254 documents, extracting 2,477 question/answer segments with schema checks, source hashes, and evidence links.
- Integrated local Ollama/Qwen models with checks for source quotes, word limits, and preservation of numbers, dates, and names; kept generated revisions subject to human review.

## Skills and Interests

- Programming: Python (Pandas, NumPy, scikit-learn, Selenium, PyTorch, TensorFlow, PySpark), SQL, R, Java, D3.js
- Software: MySQL, SQLite, Azure, PowerBI, Tableau, Snowflake, Git, Excel, PageRank, OpenRefine, AWS, GCP
- Modeling: Machine Learning, Deep Learning, Regression Analysis, Time Series Regression, Tree-Based Models
- Analytical: Data Visualization, Relational Databases, Feature Engineering, Experimental Design, A/B Testing